Scalability & Future Plan

Date	21 October 2026
Team ID	VAI-AI-2026
Project Name	VoyageAI — Personalized AI Travel Itinerary Planner
Maximum Marks	1 Mark

Future Growth & Scaling Roadmap:

Outlines horizontal scaling strategies, performance enhancements, and future commercial integrations.

Roadmap Horizon	Strategic Initiative	Technical Architecture & Goal
Phase 1 (Immediate)	Redis Response Caching	Cache popular multi-day destinations via SHA-256 hashes to deliver sub-100ms responses.
Phase 2 (Medium-Term)	Live Flight & Hotel Booking	Integrate Amadeus / Sabre GDS APIs for direct in-app flight price comparison and booking.
Phase 3 (Long-Term)	Real-Time Collaboration	WebSockets & Supabase Realtime enabling multiple friends to co-edit itineraries simultaneously.
Phase 4 (Enterprise)	Multi-Region Edge Clusters	Deploy containerized FastAPI microservices across US, EU, and APAC AWS clusters.